

Co-located Mixed Reality Classrooms: Four Case Studies from Higher Education

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ABSTRACT

Multi-user mixed reality (MR) holds great potential for classroom based teaching, but what are the most effective teaching experiences, formats and pedagogical techniques for this emerging technology? Through a series of case studies involving four professors and their students, we explore the pedagogical potential of headset-based co-located MR. Using a custom platform, a series of immersive lectures were developed and delivered covering a range of themes, each including a structured design process, live immersive lectures with students and post-lesson interviews and analysis. From these experiments emerged a common set of educational possibilities offered by the medium, including spatiotemporal narratives, direct and joint attention, virtual travel, collaborative interaction and spatial thinking. Key challenges facing the adoption of MR in educational settings were also identified. This work demonstrates an effective method for exploring the possibilities of MR and provides insights for educators, researchers and developers interested in integrating MR into educational settings.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**; **Mixed / augmented reality**; • **Applied computing** → **Interactive learning environments**; **Collaborative learning**.

KEYWORDS

Multi-User Mixed Reality, Mixed Reality Classrooms, Augmented and Virtual Reality, Case Study, Higher Education

ACM Reference Format:

Krishnan Chandran, Paschalis Dimitriou, Lorenza Wand, and Matthew McGinity. 2024. Co-located Mixed Reality Classrooms: Four Case Studies from Higher Education. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650790>

Honolulu, HI, USA. ACM, New York, NY, USA, Article 111, 7 pages. <https://doi.org/10.1145/3613905.3650790>

1 INTRODUCTION AND RELATED WORK

Collaborative MR technologies are increasingly being recognised for their potential in education, offering new forms of immersive learning experiences [2, 18, 21]. Although Virtual Reality (VR) and Augmented Reality (AR) have been extensively studied in educational settings [10], the use of co-located MR, where users wearing head-mounted displays (HMD) share the same physical space while interacting with shared virtual content, has been relatively limited. This may be due to several challenges in integrating Multi-User Mixed Reality (MUMR) into classrooms. However, with the recent availability of low-cost MR headsets such as the Meta Quest¹, it is now possible to have low-cost MUMR experiences on consumer-grade headsets [25]. The concept of co-located MR can vary among different sources [27]. For this paper, we define co-located MUMR to be multiple individuals in the same physical location engaging with the same, shared virtual and real content through mixed-reality head-mounted displays.

1.1 Trends in Mixed Reality Education

From early studies by Billingham et al. [5] to more recent works such as [26], it has been shown that MR systems can enable students to engage in shared experiences, facilitating collaborative problem-solving and collective learning. Recent pilot studies conducted in controlled environments have shown promising potential for VR and AR in education. These studies have demonstrated that VR/AR can have a positive impact on learning outcomes, motivation, and engagement [2–4, 21]. Garzon highlights the growing trend of integrating education and MR in recent publications, underscoring the importance of addressing the associated challenges and opportunities [12]. The emergence of new MR headsets, such as the Meta Quest and the upcoming Apple Vision Pro², is expected to lead to an increase in the availability and quality of consumer-grade MR headsets [16]. This, in turn, could facilitate their eventual adoption in mainstream education.

¹<https://www.meta.com/de/en/quest/>

²<https://www.apple.com/apple-vision-pro/>

1.2 Co-located Multi-user Head-mounted Mixed Reality

In the specific subclass that pertains to our study (co-located HMD-based MUMR), the scarcity of literature could be attributed to factors such as cost, technical challenges in achieving co-located MR and other practical hurdles. In the landscape of scholarly literature, a multitude of examples showcase MR-based collaboration; nonetheless, as it can be seen in Feng et al.'s [10] review, attempts to integrate MUMR to facilitate a co-located MR classroom are limited. There exist works that explore calibration strategies to align dissimilar spaces [13, 35, 35]. Xu et al., [33] propose a work which could be classified as an example of the class of systems that we are interested in. At present, only a limited number of systems offer ready-made solutions for the technical requisites of co-located MR, as exemplified by the findings of Schier et al. ([25]).

1.3 Challenges and Opportunities in Mixed Reality Classrooms

The literature contains several position papers outlining the challenges and opportunities of incorporating MR technology into education. According to Dengel et al., the primary challenge in navigating this literature is the lack of a common language regarding educational perspectives [8]. These perspectives range from pedagogical frameworks (theoretical) to design guidelines (technical).

Popescu et al. [22] identify various challenges in implementing a MR classroom for multiple users. These challenges include co-locating students in the virtual world, scene geometry knowledge, user interface design, safety, networking requirements, implementation cost, and digital content creation.

As the authors state in [2, 9], numerous pilot projects observe the impacts of educational immersive media as isolated experiences and integration of such media into the regular classroom and curriculum would require much more meticulous planning. The design and development recommendations for HMD-based VR classrooms [14, 15], also apply to HMD-based MR classrooms. Yepes et al. [34], highlight the importance of MR-specific educational design, offering guidelines for multimedia, modality, and information coherence. While not focused on co-located MR classrooms, they underscore communication support as a vital MR learning feature. The hindrance to natural communication posed by HMDs questions whether the trade-offs associated with co-located MR are substantial enough to justify their acceptance. Lastly, the lack of interface design standards and authoring tools [36] makes it difficult to compare diverse results. As Ali et al. [2] report, there is a scarcity in the number of research articles that describe experiments and experiences that have taken place in real-world settings.

1.4 Research Questions

This text presents a series of case studies involving university professors and researchers from diverse disciplines. The aim is to share insights into the technical and non-technical challenges that remain to be addressed. Additionally, potential opportunities that co-located MR presents are identified. Four university lecturers from a variety of domains were selected and invited to deliver a lecture in our internally developed co-located MR system. This text outlines the process of identifying lecture themes, collaboratively

designing the lecture, and sharing the outcomes. The findings from 7+ hours of focus group interviews are also presented.

More specifically we ask the following:

RQ1: What are the pedagogical opportunities of MR classrooms?

RQ2: What are the challenges in the integration of co-located MR into classrooms?

Our study contributes to the ongoing discourse of identifying the educational potential of MR technologies. In light of the results from the case studies, we make the following contributions.

- We present the findings from all four case studies which involved the design and development of an experimental MR lecture in collaboration with university professors.
- We identify the prevailing challenges and the potential opportunities of co-located MR.

2 METHODOLOGY

To identify the key challenges and opportunities of MR classrooms, we consider it important to conduct trials with practising teachers and students using teaching content from their syllabi. A series of case studies were conducted, with practicing university professors (T1-T4) giving immersive lectures to active students (P1-P19) with varying levels of academic proficiency (Bachelor to Doctoral candidates).

The selection of topics for experimental lectures were guided by factors such as the professors' availability, the scarcity of published examples [3, 4], the topics suitability for spatial exploration, and the professors' willingness to employ MR for teaching.

A digital twin was made for an 11 m by 12 m laboratory room. A handful of demonstrative learning scenarios were prepared using the *Unity Multi-User Mixed Reality framework ViewR* [25]. The instructors (T1-T4), used a Meta Quest Pro and the students (P1-P19), used Meta Quest 2 for the experimental lectures.

Each case study followed a five-stage protocol:

- *Step 1:* During the familiarisation session, instructors (identified as T1 to T4) were introduced to co-located MR. They were asked to provide their initial impressions and expectations. To ensure efficient use of time, a strict time budget was allocated to avoid excessive development.
- *Step 2:* Then, we conducted a two-hour long participatory design session with the instructors, where we ideated the lecture. We wanted to build an experimental lecture drawn from their syllabi and not create an entirely new lecture from scratch. During these sessions, the authors discussed the pedagogical possibilities of MR with the instructor. The outcomes of these sessions were a series of sketches (Figure 1), a storyboard and a spatio-temporal layout (Figure 2) to sequence and pre-visualise the lecture. *Storyboards, sketches and spatio-temporal layouts from all the lectures are submitted as supplementary material.*
- *Step 3:* We set ourselves a hard constraint of one week for content development with a team of two developers and one 3D modeller. The time constraint was set to simulate realistic time conditions. In practice, lecturers often have very little time to prepare for the lectures.

<u>Lecture</u>	<u>Instructor</u>	<u>Duration</u>	<u>Lecturer</u>	<u>Theme</u>	<u>No. of Students</u>
Language and Space	T1	45 min	Professor in German Linguistics and Language History	The relationship between language and spatial reasoning	five
History of Cartography	T2	45 min	Professor of Cartographic Communication	Evolution of mapmaking techniques	five
Design of Learning Environments	T3	30 min	Professor in Proto-Vocational Education	Physiological & psychological factors in designing effective learning spaces	four
Understanding Landforms	T4	45 min	Scientific Assistant and Senior Researcher in Cartography	Using mapping, imagery, and virtual models to understand geomorphology	five

Table 1: The lectures that were selected for our case study.

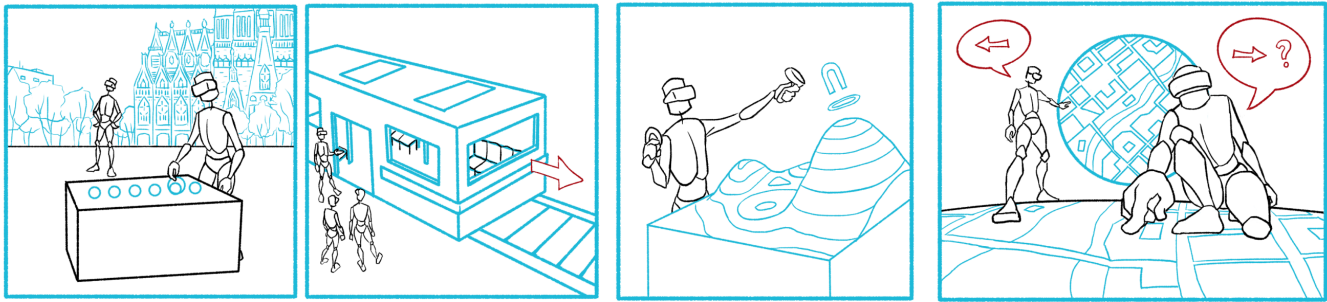


Figure 1: Selected sketches from the participatory design sessions. The goal of these sketches was to pre-visualise the MR pedagogical exercise before the implementation. Interactive Panorama Explorer from T2's lecture (leftmost). Virtual train journey from T3's lecture (center-left). Virtual sculpting tool from T4's lecture. (center-right). Spatial orientation exercises on floor maps from T1's lecture (rightmost). The full set of sketches are available as supplementary material.

- *Step 4:* During the following session, the instructor presented a lecture on MR to a selected group of students. The lecture was recorded using video, audio, and MR captures. In addition, a camera operator attended the lecture as a non-participant to mirror the MR view from the headset. The exact duration of the lecture varied (30-45 min) and is described in the case study section.
- *Step 5:* Immediately after the lecture, the students were handed a questionnaire. (The questionnaire used for the evaluation is presented as supplementary material). We then conducted focus group interviews based on the guidelines provided by Venkatesh et al. [32] with the instructors and the students in a hybrid of German and English. The furniture was arranged circularly to facilitate more fluid conversations. The audio recordings from the group discussions were later transferred to the computer for the literal transcription. The lecture titles and descriptions can be found in Figure 1.

The lecture "Language and Space" explored the relation between language and spatial orientation through collaborative tasks. Drawing on a 2021 lecture series offered by T1, students engaged in spatial orientation experiments combining pointing and linguistic gestures to foster cognitive alignment and perspective-taking. In the "History of Cartography" lecture, the instructor incorporated map interpretation exercises, and interactive tools for concepts like circumference calculation and Mercator projection, and concluded with a 360-degree panorama to showcase the temporal evolution of cartography. The contents were based on a university lecture offered by T2. The lecture "Design of Learning Environments", delved into the physiological and psychological factors of learning environment design using animated slideshows, 3D models, and a virtual train journey to illustrate context's impact on perception. This lecture drew on a university lecture offered by T3 titled *Gestaltung von Lernumgebungen: Physiologische und psychologische Grundlagen der Wahrnehmung*. The lecture "Understanding Landforms", featured a

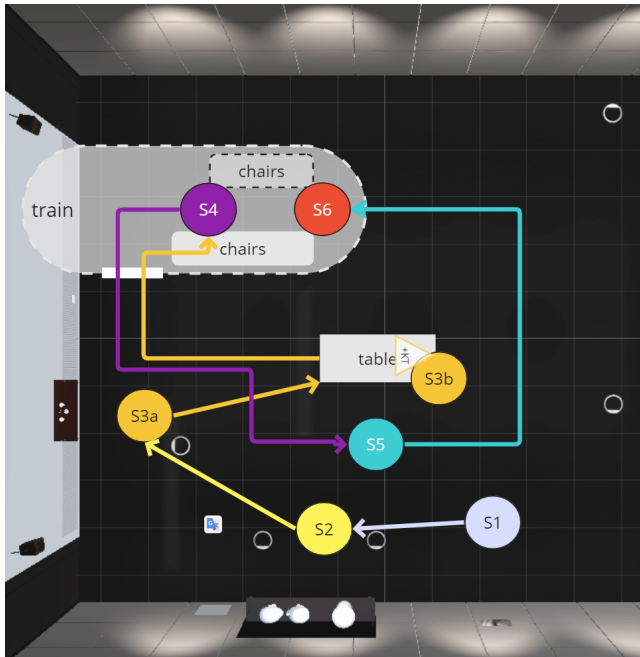


Figure 2: Spatio-temporal layout of the lecture with T3. The icon S1 denotes station 1, where the first sequence of the lecture was planned. A detailed storyboard was also made which denotes what happens at each station, this includes the positioning and activities of the teacher and the students. Storyboards and Layouts from the other lectures are included in the supplementary material

field trip with a triplet matching task and a virtual landscape sculpting tool to enhance skills in detecting and understanding landforms while exploring their morphological feasibility. This was based on the field trips that T4 offered every semester for Geoinformatics students.

3 RESULTS AND ANALYSIS

3.1 Questionnaire

Participants' experiences with MR showed differing levels of physical safety. Concerns about collisions and headset comfort influenced overall user comfort. Additionally, participants reported being aware of misalignments between the real and virtual environments in the MR classroom. Responses varied based on occasional drift experienced with mobile HMDs. Although some participants were distracted by the use of MR, the majority recognised the system's effectiveness in delivering and comprehending educational content.

Open questions revealed technical issues faced by participants, such as sudden environmental changes and difficulty grabbing

virtual objects. The suggested solutions focused on device improvements. Participants envisaged future MR systems with perfect features, highlighting their potential applications in visualizing complex concepts, exploring history and facilitating collaborative annotation, and emphasising the importance of simplicity and comfort for an enhanced learning experience, similar to wearing glasses.

The questionnaire and the results can be found in the supplementary material.

3.2 Focus Group Interview

In all case studies, instructors (T1 to T4) actively engaged in discussions for 90–120 minutes per session. Additionally, 19 students (P1 to P19) contributed to the scholarly exchange, providing a wide spectrum of viewpoints and experiences. The group discussed several themes, including a detailed conversation about their vision for an ideal MR classroom. We also discussed the major challenges we encountered during the session and the potential opportunities.

The instructor from Case Study 1 described how MR could be used to direct and control students' attention towards specific learning content during lectures.

T1: I think [...] very key about this [MR classroom is], it involves directing your attention. I think that is a skill that is sometimes hard to teach in a class.

Referring to contextual shifts to demonstrate didactic concepts, T2 describes the following:

T2: The teacher can maybe rate it [a certain learning experience] differently because you have a completely different perspective. I think it's interesting to play with this perspective. I think [from] the other direction from the student's perspective or the instructor's perspective [...], the student sees even more and has expanded the experience.

Additionally, interactive instruction such as virtual modelling could also greatly benefit the understanding of concepts as was seen in Case Study 4. Participants (P16–P19) expressed that the virtual sculpting of terrain greatly helped in understanding the morphological possibility of the user-sculpted terrain and the evolutionary processes behind it.

T4: And I think the sculpting is superb, then the students could model different situations and say, "I recognize", for example, "the alluvial fan" or so, "in this real-world model at this spot" or so, and they maybe could draw a little outline on that.

Referring to the spatial thinking affordance of MR, T1 described the following:

T1: Language is deeper and that has different levels on which, for example, gamification of the morphology, syntax, and so on, and tree structures. You can give the red cube the meaning of 'Baum' and the yellow one of 'loud,' and then give a sentence and say, "Build the sentence with this cube."

We explain the results from the interview further in the discussion section.

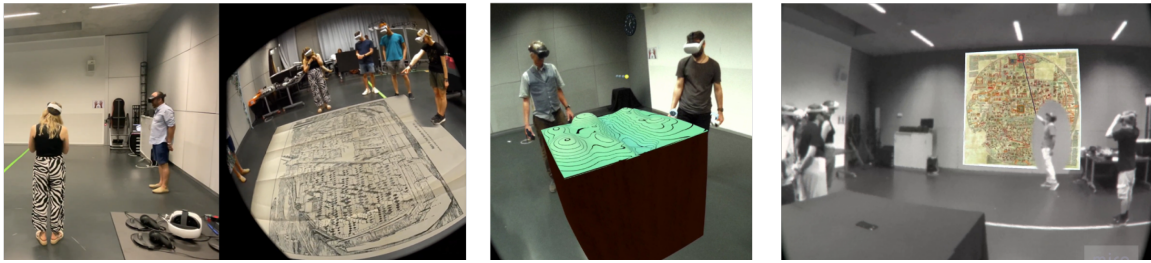


Figure 3: Selected screenshots from the lectures (MR view captured from the HMD). Historical City Map laid out on the floor as viewed by T1 through the HMD, from the lecture "Language and Space" (*left*). Interactive Virtual Sculpting from the lecture "Understanding Landforms" (*middle*). T2 presenting the Ebstorf map in the lecture "History of Cartography" (*right*).

4 DISCUSSION

4.1 Pedagogical Opportunities in Mixed Reality classrooms

From the results of the case studies, we observe the following pedagogical opportunities emerge.

4.1.1 Directing and Joint Attention. As the instructor from Case Study 1 described, MR could be used to direct and control students' attention towards specific learning content during lectures. While the idea of guiding attention has been explored in cinematic VR, it has not yet been applied in educational settings [20]. Efficiently controlling and directing the attention of multiple students is still an open question.

4.1.2 Collaboration. Collaborative annotation is another interesting area of research. MR collaborative tools exist for annotations such as virtual hand gestures, ray pointing and drawing annotations [29], but the best interfaces for text annotation are still an open problem. An additional feature of interest to the lecturer in Case Study 2 is the ability to have individual lecture notes. A persistent role-based note-taking system could be a valuable addition to the MR classroom.

4.1.3 Virtual Travel and Contextual Shifts. Educational lessons that use contextual shifts to demonstrate didactic concepts could greatly benefit from a MR system, as shown in Case Study 2. Similarly, educational applications that emphasize a change in perspective can also benefit from such a system, allowing learners to adopt multiple vantage points and expand their educational experience.

4.1.4 Interactive Concept Visualisation and Understanding. Additionally, interactive instruction such as virtual modelling could also greatly benefit the understanding of concepts as was seen in Case Study 4. Such interactive visualisations and simulations can facilitate inquiry-based learning [28] by actively exploring "What-if" scenarios. Other domains such as molecular modelling, or engineering prototyping could also greatly benefit from such interactive simulations in the classroom.

4.1.5 Spatial Thinking. As demonstrated in Case Study 1, MR can be used to represent abstract concepts such as language. It is worth exploring the potential of spatial visualisation for other abstract concepts, such as *time* and *relationships*. The field of immersive analytics has seen significant advancements [11], and integrating

these tools into the MR classroom could offer valuable instructional and educational opportunities.

4.1.6 Student Monitoring. Moreover, the capacity to monitor student data offers a chance to incorporate student engagement metrics into our educational practices. It is important to note that progress has been made in this area [1]; however, obtaining more comprehensive engagement metrics from a wider range of data sources remains a challenge.

4.1.7 Other Opportunities. Moreover, the capacity to monitor student data offers a chance to incorporate student engagement metrics into our educational practices. It is important to note that progress has been made in this area [1]; however, obtaining more comprehensive engagement metrics from a wider range of data sources remains a challenge. Similar to Nebeling et al.'s findings, our case studies suggest that complex concepts benefit most when expressed spatially and sequentially in a co-located MR classroom [19].

4.2 Spatio-Temporal Presentations

Each lecture utilised space in unique ways for delivering the lecture, in a variety of a "Gallery walk" [31] patterns (T1-T4). Additionally, instructors also used space to move through abstract information such as language and cartographic data (T1-T2). Furthermore, T4 used the space to design virtual field trips and *treasure hunts*. Alluding to such "immersive" presentations, the need for a tool to orchestrate the MR classroom was a common sentiment across the interviews. Conventional graphical presentation tools which organise the content in the form of slides force the teacher to structure their class linearly and rely on a hierarchical single-path structure as a model for the organization, regardless of the content [30]. This rigidity might be alleviated with the spatial dimensionality of MUMR. The expanded metaphors, ranging from 3D slideshows to gamified elements, offer educators diverse ways to present information, fostering collaboration and exploration in persistent spatial knowledge spaces. Immersive spaces are already being used for sensemaking[7], and these strategies could also be used as didactic devices in MR classrooms.

4.3 Challenges

4.3.1 Content Creation and Implementation. Content creation remains a significant challenge, requiring substantial development

efforts. Technical developers must collaborate with educators possessing pedagogical expertise. Other impediments include a lack of common vocabulary, limited educator awareness of possibilities, and misjudgment of required effort. Educators face challenges due to their busy schedules, which limit their time to prepare lectures. To address this, we suggest leveraging existing content and reusing the educator's knowledge base. Another challenge is to determine the ability of educators to acquire and integrate immersive content such as volumetric videos, 360-degree videos and point clouds into MR classrooms easily. [24, 35] Additionally, we propose the adoption of guidelines for managing responsibilities within the MR classroom environment [15, 26]. Notably, the challenges we identified during our case studies mirrored those found in broader XR collaboration research [17].

4.3.2 User Experience. If mobile MR headsets are used in the classroom, it is important to consider their rendering limitations. Furthermore, subjects that heavily rely on textual data, such as language and history, may require alternative methods for representing this information in MR.

A recurring issue we noticed is the repetitive use of phrases, such as "Are you seeing what I am seeing?" This could mean that the users are unsure of what the other users are seeing. This a challenge, and alternative methods of conveying the user's perspective, such as gaze pointers, should be considered. Another challenge is maintaining student engagement throughout the lecture. Furthermore, there is a need for more precise user representations. Techniques such as passthrough avatars [25] can obscure virtual content, making intelligent keying necessary.

4.4 Limitations

We acknowledge the following limitations of our case studies. It should be noted that the topics covered in the lectures are not exhaustive, and more variety of topics could be explored. Currently, we are exploring professors from domains where there is less published work with MR lectures. It is important to consider the teacher's behavioural profile in future case studies. Our study found that teachers need to be highly tolerant of unexpected issues during the lecture. To extend the applicability of our findings, we intend to conduct a follow-up study with a larger sample size. Although the teachers wore a Quest Pro in our instructor-focused case study, the students wore a Quest 2, which is no longer state of the art and did not have colour passthrough, this may have had an impact on student learning. This warrants further investigation. This study focused solely on visual elements, with limited exploration of haptics beyond basic objects like desks/boxes (passive haptics). Future research could be highly beneficial in investigating tangible user interactions and their pedagogical affordances in co-located MR classrooms [6, 23].

5 CONCLUSION AND FUTURE WORK

Our work is the first step in validating the positions put forward by Popescu et al. [22] and highlights the potential of MR classrooms in universities. We conducted four case studies with practising university professors and identified the pedagogical opportunities in co-located MR classrooms. We categorise the opportunities as directing & joint attention, spatiotemporal narratives, virtual travel,

spatial thinking, interactivity and collaboration. Furthermore, we outline the major challenges to MR content creation, workload management, student engagement and interaction challenges. Our future work involves identifying the design requirements and the development of a general-purpose presentation software suite for MR that could enhance MR classrooms.

ACKNOWLEDGMENTS

The authors would like to thank Mucella Cetinkaya for the sketches during the design session. The authors would also like to thank Dr Nicholas Prechtel, Prof Dirk Burghardt, Prof Alexander Lasch and Prof Rolf Koerber for playing the role of teachers in the lectures and special thanks to all the students who took part in the immersive lectures. We also thank Daniel Zeidler and Victor Victor for their technical assistance during the case studies. We thank Julius Heinkauf for his assistance during the semi-structured interview.

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